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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Automation Engineer with 3 years of industry experience at HDFC Asset Management delivering production automation and AI inside real business workflows for the Compliance, Client Service, Investment and Sales teams: a Python pipeline that parses 300+ rating-agency emails a day (saving 60 analyst hours a month), Azure OpenAI tool-calling agents including a WhatsApp fund-comparison bot used daily by roughly 50 to 100 Sales team members, and RAG applications on PostgreSQL with pgvector, integrated through APIs, scheduled events and company authentication. First engineer on the GenAI systems alongside my manager, working directly with the business users; later de facto team lead for up to 4 interns; Best Team Project twice. MSc in Data Science at the University of Bristol (awaiting result). Seeking an AI automation role that removes manual effort from complex financial-services processes.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production AI Automation in Business Workflows

- Built a **Python** automation pipeline that parses over **300 daily emails** from credit rating agencies, fuzzy-matches them to the companies analysts track (dropping the rest) and sends a daily consolidated email of previous and changed ratings, **saving 60 analyst hours a month**; run on **AWS SageMaker** and scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM. Replaced **AWS Textract** with **Azure OpenAI** when one agency began sending its emails as images, which also improved accuracy.
- Built an **agentic WhatsApp fund-comparison bot** (via the **Netcore** API) used daily by the Sales team (roughly 50 to 100 users, 400 to 1,000 requests in working hours): **Azure OpenAI tool calling** picks the comparison type and funds and calls **Python functions** over SQL data, returning fund indicators and a chart PDF in the chat. Parameter extraction broke first; fixed with example inputs in the JSON tool definitions.
- Designed and built **RAG** applications end to end for Compliance, Client Service and Investment (**ReactJS**, **Python FastAPI** microservices, **LlamaIndex**, **Azure OpenAI GPT-3.5 & Llama-3**), with a separate **pgvector** database per department to keep access siloed; guardrailed behaviour through **system prompts**, versioned every prompt change in **Git** and had Compliance review each release.

Data Integrations & Business Users

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Integrated the automation with the tools the business already used: **WhatsApp** and a shared **email** inbox as the interfaces, the **Morningstar API** and **PostgreSQL** as data sources, and company **LDAP** for user authentication, with the same SQL data behind the bot, the dashboards and the RAG applications.
- Developed **Power BI** dashboards for the investment team's credit report analysis, moving analysts from Excel and improving efficiency by **10 hours a week**; recreated them in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs.

- Worked directly with Investment, Compliance and Client Service stakeholders; **de facto team lead for up to 4 interns**; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

Deployment, Security & Reliability

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** (**EC2 including a GPU instance, S3, ECR and Lambda**), packaged the **FastAPI** microservices as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD** pipeline.
- Implemented **RBAC** with **LDAP** authentication, restricted access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: the **RAG-based AI assistant** that narrates ranked results and answers methodology questions; the **scoring and ranking algorithm** across six evidence layers; and the **data preprocessing pipeline** (14 clean, joined tables from 17 raw inputs, with integrity checks). Returns a ranked shortlist of cancer cell lines with the evidence behind each pick, refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with **108 LLM personas** across 12 archetypes and a vision model that scores uploaded photos (ReactJS, Python FastAPI, NVIDIA NIM with the model selectable per request). LLM output treated as untrusted: **validated with Pydantic, repaired and cached by input hash**; rate limits shaped the design (12 calls instead of 108, results streamed). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. **Multi-architecture deep learning study** comparing CNN, ResNet and ViT encoders trained from scratch with LSTM/attention decoders on three synthetic datasets across five seeds, evaluated with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle credit card fraud detection problem.)
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework for AI-assisted coding) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU, S3, ECR, Lambda, SageMaker, EventBridge), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite